

# Deep Reinforcement Learning

## *Introduction*

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# Who am I?



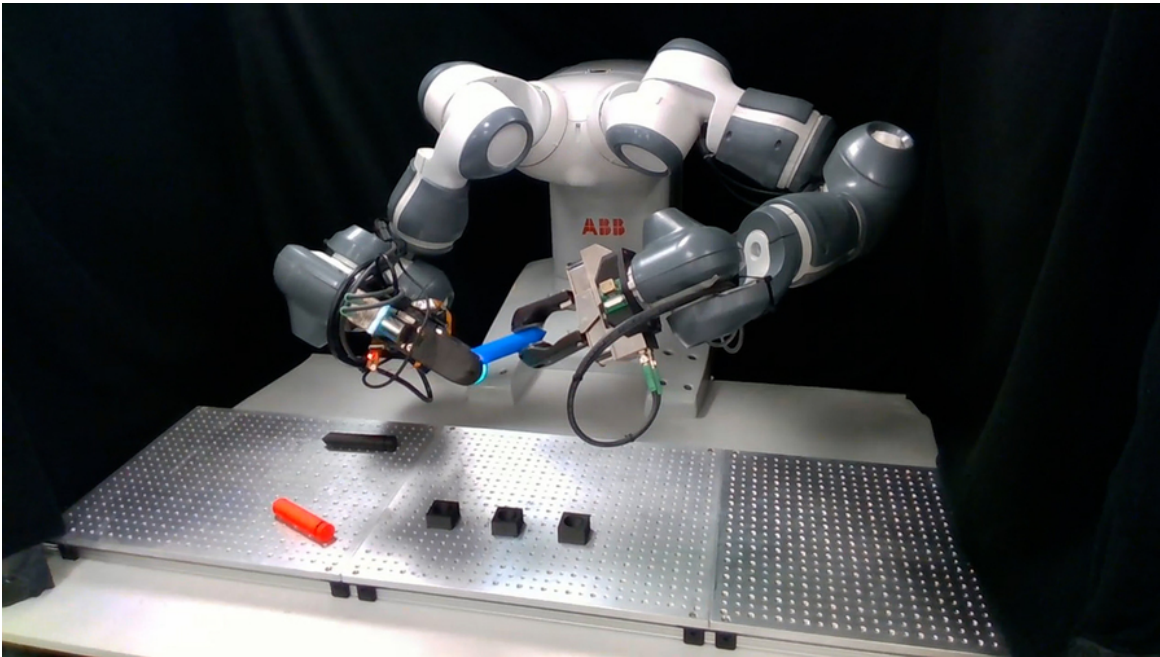
- Prof. Dr. Sebastian Peitz
- Chair of Safe Autonomous Systems
- JvF25, Room E16
- <https://sas.cs.tu-dortmund.de/>

# The team

- Lecture: Sebastian Peitz
- Exercise: Konstantin Sonntag
- Additional team members working on the exercises
  - Jannis Becktepe
  - Septimus Boshoff
  - Hans Harder
  - Christian Mugisho Zagabe
  - Stefan Weigand

# Contents of the Lecture

# Sequential decision making: Taking actions that affect our environment



Task: pick up pens and place them upright [Source]

## Option 1: Rule-based system

- manual design of actions, given some sensor input (such as a camera)

## Option 2: Model-based control

- given some model of the dynamics, optimize for the best possible action to take

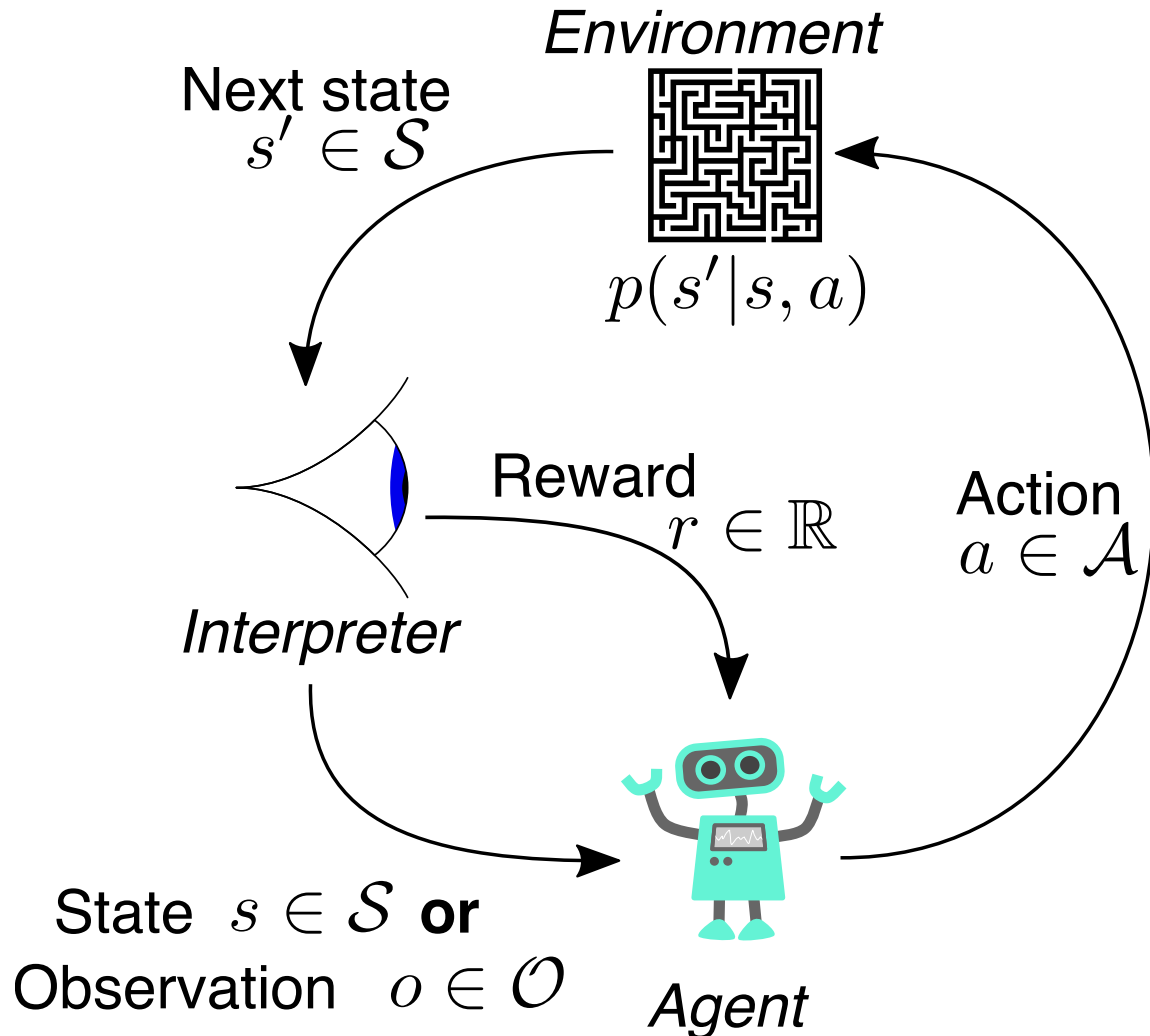
## Option 3: Imitation learning

- collect data from experts and try to *clone the behavior*

## Option 4: Reinforcement learning

- collect data in a *trial-and-error* fashion, improve via feedback

# What is reinforcement learning?

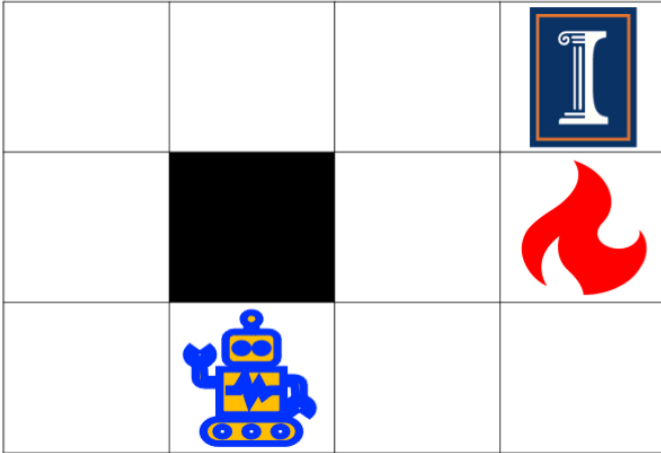


## In words

- An **agent** perceives the state  $s$  of its environment.
- It takes an **action**  $a$  according to a **policy**  $\pi(a|s)$ .
- The environment changes due to the action:  $s \rightarrow s'$ .
- The agent receives a **reward**  $r$ .

## The goal

# Reinforcement learning examples (1)



*Grid world*

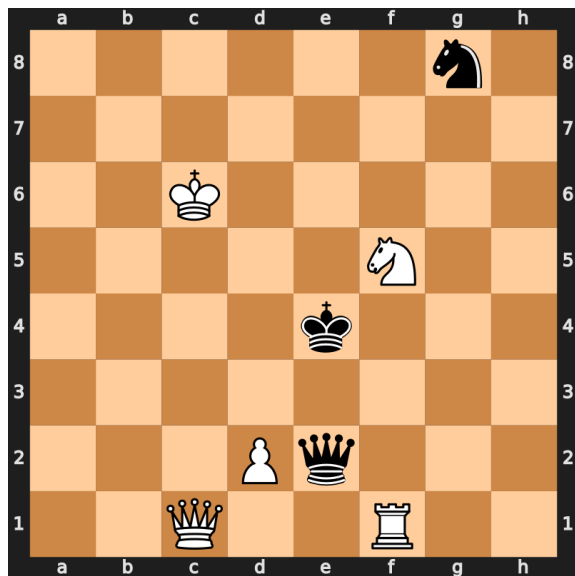
- **Environment:** The grid world
- **Agent:** The robot
- **State  $s$ :** The robot's position
- **Action  $a$ :** Move left / right / up / down
- **Dynamics  $p(s'|s, a)$ :** The robot's new position...
  - ...deterministically, if the robot does exactly what the action demands
  - ...stochastically, if there is some noise in the taken action

- **Reward:**

$$r = \begin{cases} +1 & \text{if you reach the target field (top right)} \\ -1 & \text{if you hit the fire} \\ -0.1 & \text{otherwise} \end{cases}$$

- **Policy  $\pi$ :** The strategy according to which you move

# Reinforcement learning examples (2)



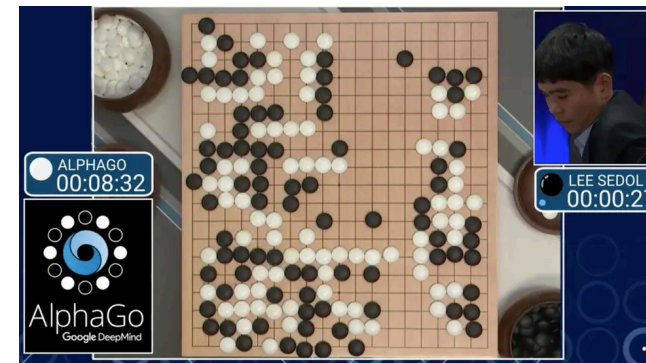
Chess board [Source]

- **Environment:** The chess board
- **Agent:** The chess player(s)
- **State  $s$ :** The board position
- **Action  $a$ :** Move one of your figures
- **Dynamics  $p(s'|s, a)$ :** The new board position...
  - ...after your own and your opponent's moves → your turn again
  - ...after your move; it's the other players turn → two-agent game

- **Reward:**

$$r = \begin{cases} +1 & \text{if you take your opponent's king} \\ -1 & \text{if your king is taken} \\ 0 & \text{otherwise} \end{cases}$$

- **Policy  $\pi$ :** The strategy with which you play



AlphaGo (Silver et al. 2016) [Source]

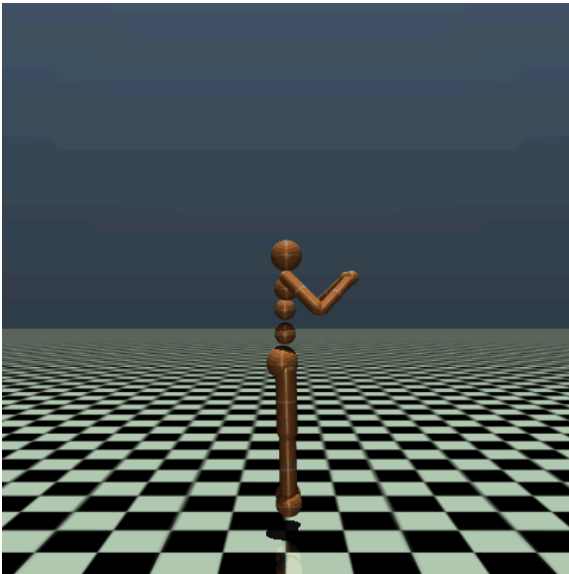
# Reinforcement learning examples (3)



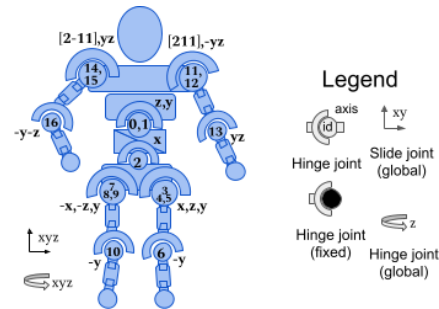
Pendulum [Source]

- **Environment:** A pendulum under gravitational force
- **Agent:** The control system trying to perform a swing-up
- **State  $s$ :**  $\sin$  and  $\cos$  of the angle  $\varphi$  (top:  $\varphi = 0$ ) and angular velocity  $\omega = \frac{d\varphi}{dt}$
- **Action  $a$ :** the applied torque
- **Dynamics  $p(s'|s, a)$ :** a differential equation describing the dynamics (*Newtonian mechanics*)  $\rightarrow$  deterministic system
- **Reward:**  $r = -(\varphi^2 + 0.1 \cdot \omega^2 + 0.001 \cdot a^2)$
- **Policy  $\pi$ :** the torque realizing the swing-up

# Reinforcement learning examples (4)



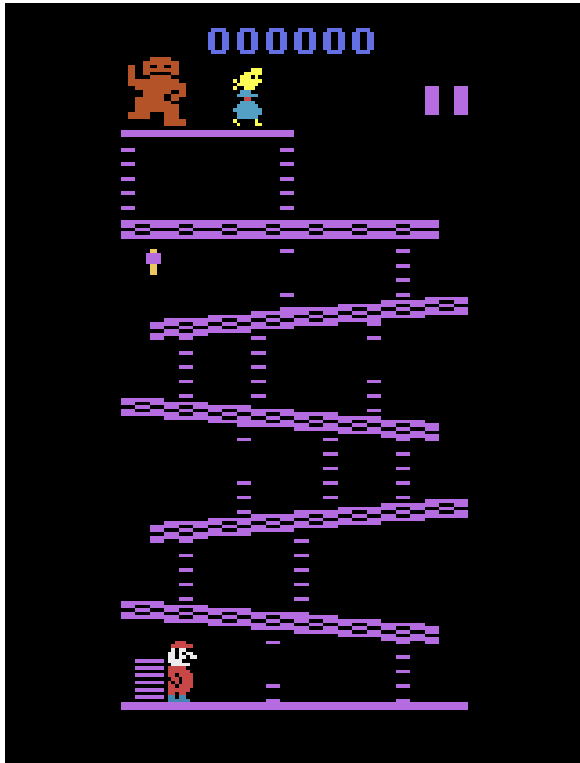
Humanoid [Source]



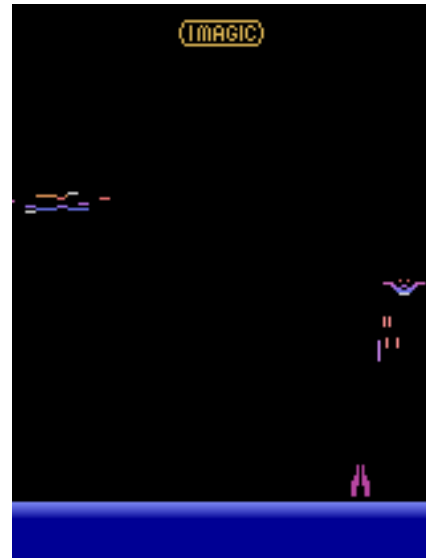
Model

- **Environment:** A robot moving on a planar ground
- **Agent:** The control system for actuating the robot
- **State  $s$ :** Positions and velocities of the components (45 in total)
- **Action  $a$ :** The torques applied to the 17 joints
- **Dynamics  $p(s'|s, a)$ :** a differential equation describing the dynamics (*Newtonian mechanics*)  $\rightarrow$  deterministic system
- **Reward:** healthy reward + forward reward - control cost - contact cost
- **Policy  $\pi$ :** Run as far as you can!

# Reinforcement learning examples (5)



Atari: Donkey Kong [Source]



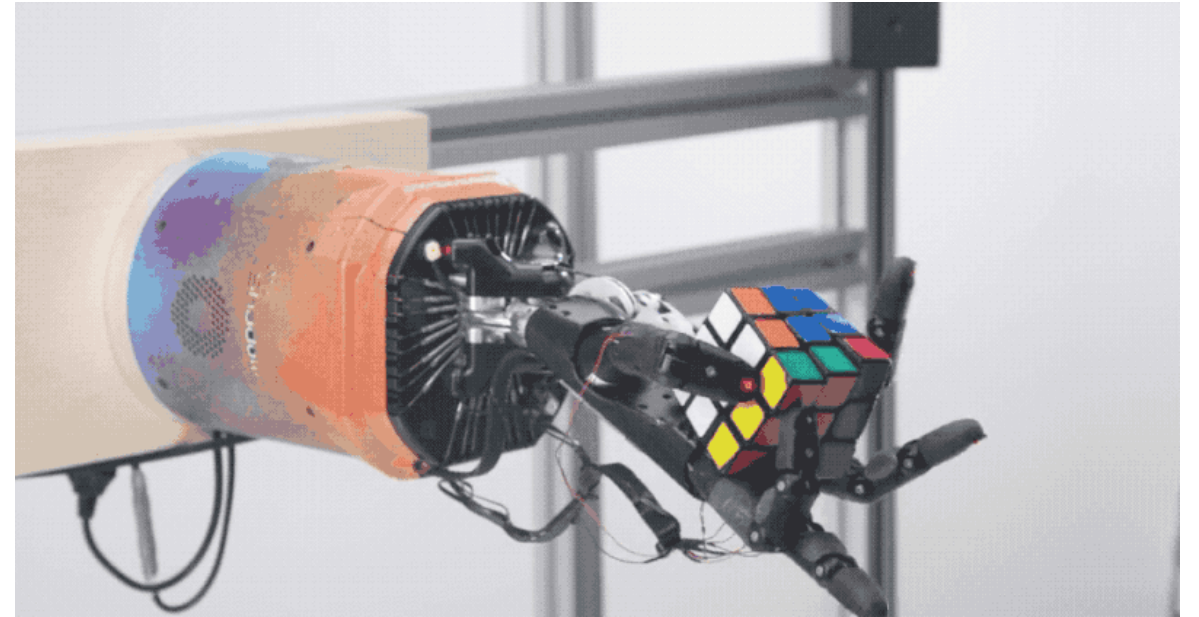
Atari: Demon Attack  
[Source]

- **Environment:** Atari video games
- **Agent:** The “player”
- **State  $s$ :** the pixels of the screen
- **Action  $a$ :** the game pad (i.e., moving, firing, etc.)
- **Dynamics  $p(s'|s, a)$ :** the video game reacting to the player's actions
- **Reward:** achieving the game's goals (stay alive, reach a target, eliminate opponents, ...)
- **Policy  $\pi$ :** play such that you achieve your goal as best as possible

# Reinforcement learning in robotics



Boston Dynamics "Atlas" [\[Source\]](#)



OpenAI Rubik's Cube [\[Source\]](#)

# The two central tasks in RL

In all chapters, we will distinguish between **two core learning tasks in reinforcement learning**:

## 1. The prediction problem (*evaluation*)

- For a fixed policy  $\pi$ , we want to **estimate the quality of a state  $s$**  (i.e., its **value**).

## 2. The control problem (*improvement*)

- We want to **improve our policy  $\pi$**  towards the optimal  $\pi^*$  that maximizes the value.
- This usually requires a back-and-forth between 1. and 2.
  - Given our estimate of the value, we improve the policy:  $\pi' > \pi$ .
  - We then need to reiterate the values given the new policy  $\pi'$  and so on.

# The key difference to supervised learning

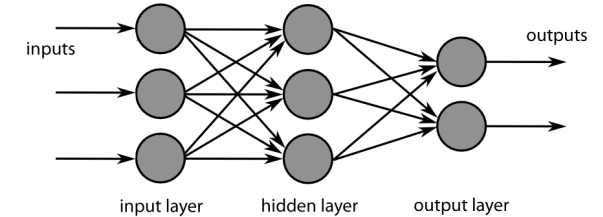
- **Supervised learning:**

- minimize error on a training dataset  $\mathcal{D} = \{(x_1, y_1), \dots, (x_K, y_k)\}$ :

$$\min_{\theta} \sum_{k=1}^K \|y_k - f_{\theta}(x_k)\|_2^2$$

- data is generally drawn independently and identically distributed (i.i.d.):

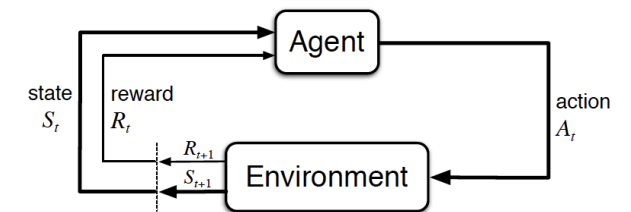
$$(x_k, y_k) \sim p(\mathcal{D}).$$



*Feed forward neural network*

- **Reinforcement learning:**

- no supervisor, just a *reward signal*
- feedback is delayed, not instantaneous
- time really matters (sequential, non i.i.d. data)



*Reinforcement learning (Sutton and Barto 1998)*

- agents actions affect the subsequent data it receives

# The big questions

- How can we learn a good policy  $\pi$  *from experience*?
- How can we address the *exploration-exploitation dilemma*?
- How do we use *deep learning* to address otherwise intractable problems?
- Which techniques can we use to facilitate *efficient and stable learning*?
- What are interesting *applications of RL*?
- What are the *open questions in RL research*?

# Goal of the lecture

By the end of the course, we will

- understand the **theoretical basis** behind RL
- know the **landscape of RL algorithms** as well as their pros and cons
- be able to **implement RL algorithms** (using python)
- be capable to **critically assess** scientific papers as well as practical RL methods
- know how to tackle **real-world problems** using RL

# Chapters

- **Introduction**
- **Multi-armed bandits**
- **Basics & tabular methods**
  - Markov Decision Processes (MDPs)
  - Dynamic Programming
  - Monte Carlo Methods
  - Temporal Difference Learning & Q-learning
- **Deep-learning-based methods**
  - Brief introduction to deep learning
  - Value function approximation
  - Deep Q-learning
- **Model-based control**
  - Planning & optimal control
  - Model-based RL
  - Exploration
- **Advanced topics**
  - Offline RL
  - Transfer learning
  - Multi-agent RL
  - Behavior cloning

- Actor-Critic algorithms
- Advanced algorithms

# Organization

# Lectures & Exercises

## Lectures

- Tuesdays (10:15 - 11:45, OH16 / 205)
- Wednesdays (08:30 - 10:00, OH12 / E.003)

## Exercises

- Thursdays (10:15 - 11:45, OH16 / 205)
- One exercise sheet per week
  - pen-and-paper exercises the cover and extend the theoretical content of the lecture
  - programming exercises (in Python) supporting the Studienleistung (next slide)

# Studienleistung – Programming tasks

- over the semester, we will publish four long-term programming tasks
  - they cover 3-4 worth of content weeks
  - the programming exercises on the weekly exercise sheets cover parts of these programming tasks
  - to be accepted to the exam, **you need to pass at least 3/4**
- task evaluation
  - the tasks are pass/fail
  - we will take them into consideration in the final exam by dedicating several questions to your submissions
  - this way, they will account for  $\approx \frac{1}{3}$  of your final grade
- usage of ChatGPT, Gemini, Claude, etc.
  - is allowed 😊

# Exam

## Format:

- oral exam during the semester break (dates to be announced soon)
- 30 minutes per candidate

## Content:

- all topics covered in the lectures
- all topics covered in the exercises
- in-depth questions regarding your submitted programming tasks regarding
  - your choices regarding modeling / implementation
  - your results
  - your usage (e.g. prompts) of LLMs, if applicable

# Literature

## Reinforcement learning

- Reinforcement learning: an introduction (Sutton and Barto 1998)
- David Silver's RL lecture (Silver 2015)

## Mathematical basics

- Linear Algebra: Stanford CS229 review material
- Probability:
  - Stanford CS229 review material
  - Introduction to Probability lecture notes by Dimitri Bertsekas

## Programming

- Berkeley's CS285 pyTorch course ([Google Colab](#))

# References

Silver, David. 2015. “Lectures on Reinforcement Learning.” url: <https://www.davidsilver.uk/teaching/>.

Silver, David, Aja Huang, Chris J. Maddison, Arthur Guez, Laurent Sifre, George van den Driessche, Julian Schrittwieser, et al. 2016. “Mastering the Game of Go with Deep Neural Networks and Tree Search.” *Nature* 529 (7587): 484–89.

Sutton, R. S., and A. G. Barto. 1998. *Reinforcement Learning: An Introduction*. MIT Press.